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# Journal of International Money and Finance

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## The behavior of currencies during risk-off episodes<sup>☆</sup>



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### ARTICLE INFO

#### Article history:

Available online 9 January 2015

#### JEL Classification:

F3

#### Keywords:

Risk-off episodes

Exchange rates

Safe haven currencies

### ABSTRACT

Episodes of increased global risk aversion, also known as risk-off episodes, have become more frequent and severe since 2007. During these episodes, currency markets exhibit recurrent patterns, as the Japanese yen, Swiss franc, and U.S. dollar appreciate against other G-10 and emerging market currencies. The pattern of these moves can be explained by a combination of fundamental factors, such as the nominal interest rate, the international investment position and measures of exchange rate misalignment, and market-liquidity factors, such as bid-offer spreads and restrictions on international capital flows. We also find that currency performance in a risk-off episode has become more related to a currency's yield and relationship to broader risks in recent years.

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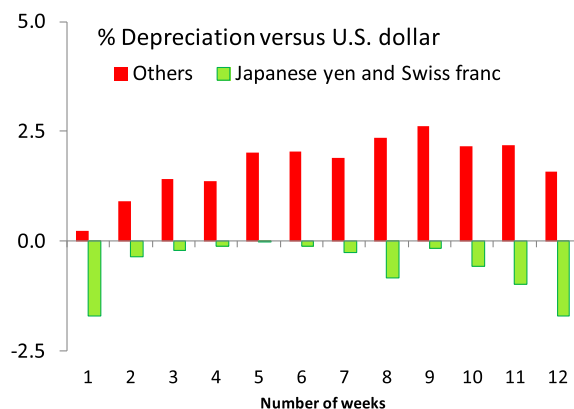
## 1. Introduction

What determines currency returns when investors across the world want to reduce their risk exposure at the same time? Most investors would agree that there are episodes – referred to in the financial press as risk-off episodes – when risky positions are cut across asset classes, as if there were a

<sup>☆</sup> We thank the editor and 2 anonymous referees for comments that substantially improved our submission. This work would not have been possible without the assistance of Sarma Jayanthi and Mauricio Vargas; and the hospitality of the Swiss National Bank. We are grateful to Steven Phillips for many stimulating discussions. Serkan Arslanalp, Lamine Bougueroua, Peter Dattels, Steven Englander, Patrick Imam, Marcel Kasumovich, Kornelius Klobucar, Mico Loretan, Michael Melvin, Evan Papageorgiou, Rodrigo Sekkel, Ivan Tchakarov, Younes Zouhar and seminar participants at the Swiss National Bank, National University of Singapore and IMF also provided helpful comments. Remaining errors are ours.

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**Fig. 1.** FX spot returns at different horizons, average of 8 risk-off episodes. Note: Others is the equally-weighted average USD return on the currencies of Argentina, Australia, Brazil, Canada, Chile, China, Colombia, Czech Republic, Denmark, Egypt, Euro area, Hungary, India, Indonesia, Israel, Republic of Korea, Malaysia, Morocco, Mexico, New Zealand, Norway, Peru, Philippines, Poland, Romania, Russian Federation, Singapore, South Africa, Sweden, Thailand, Turkey, Ukraine, United Kingdom and Venezuela. We define the start of a risk-off episode when the VIX moves 10 percentage points above its 60-day backward-looking moving average.

Source: Bloomberg and authors' calculations

rise in risk aversion globally. As such, risk-off episodes undermine the benefits of diversification within and across asset classes.

In this paper, we identify risk-off episodes based on the VIX, a measure of U.S. equity market volatility that is widely-used as a proxy for global risk aversion.<sup>1</sup> The VIX has several desirable properties for our analysis. First, it can be measured at a high frequency and in real time, so tracking the index can help to identify the onset of risk-off episodes. Second, since the VIX is derived from options prices on the S&P 500, there is no immediate link with FX markets. Finally, risk-off episodes derived from this measure correspond to our priors about disruptive market events in the last two decades.

Using this gauge for risk-off events, we first note that currency markets exhibit recurrent patterns during risk-off episodes, as the Japanese yen, Swiss franc, and U.S. dollar (USD) tend to appreciate against other G-10 and emerging market currencies, be it 1-week or 12-weeks after the start of a risk-off episode (Fig. 1).

In particular, returns of emerging markets (EM) currencies vis-à-vis the USD exhibit a high degree of correlation during and across these episodes. In this paper we show that these recurrent patterns of FX returns in risk-off episodes can be identified by simple correlations, as the cross section of spot returns against the USD is positively correlated across episodes, but also by Vector Auto Regressions (VARs).

We also find that the cross-section behavior of bilateral exchange rates with respect to the USD can be linked to information on policy interest rates and balance of payments dynamics available *prior* to the risk-off episode. The predictability of spot returns conditional on entering a risk-off episode suggests that two non-exclusive mechanisms may be at play. Investors may reassess the riskiness of each currency in light of a changing global environment (new information) or reprice risky assets in general (new price of risk).

Our research shows that lower policy rates, stronger current account balances, and stronger net foreign asset or reserve asset positions are factors related to smaller risk-off depreciations or larger appreciations. Market characteristics also help to explain FX performance during risk-off episodes,

<sup>1</sup> The VIX is derived from options prices on the S&P 500 index, and informs us about volatility and risk pricing in the US equity market. The VIX is often given a broader interpretation, as it is highly correlated with broader measures of financial stress (such as the Financial Stress Index developed by the Federal Reserve Bank of St. Louis) and with bond market indicators, including spreads on sovereign bonds of emerging market countries.

as less liquid currencies, as measured by bid-ask spreads, tend to weaken more during risk-off episodes.<sup>2</sup>

In addition, market prices and positioning prior to the onset of risk-off episodes help predict currency behavior in a risk-off episode. Currencies with a higher beta with the Japanese yen per Australian dollar (AUDJPY) are also prone to sell-offs. This beta gives a sense of how a given currency trades relative to the highly liquid AUDJPY pair that is popular with investors looking to implement G-10 carry trades. In addition, option prices are informative as well. Currencies are less resilient during risk-off episodes if at-the-money options are more expensive or investors are willing to pay more to insure against sharp currency swings.

Finally, we examine how responses to risk-off episodes have changed over time. Risk-off episodes have become more frequent and severe since the beginning of the Global Financial Crisis in the summer of 2007, after a few years of relatively calm markets. We find no substantial difference over time in the role of fundamentals associated with the balance of payments or international investment position. But a currency's yield has become a better predictor of risk-off depreciations during more recent risk-off episodes.

This paper has a unique focus on currency behavior across risk-off episodes but our results are related to an extensive recent literature. Brunnermeier et al. (2008) link the crash risk of carry trade strategies to funding constraints of leveraged investors. Rinaldo and Söderlind (2010) study the safe haven properties of the major advanced economies' currencies.<sup>3</sup> They find that the Japanese yen, and to a lesser extent, the Swiss franc display safe haven characteristics. These currencies tend to appreciate when global equity prices fall and FX volatility increases, and more so when those events are extreme. Grisse and Nitschka (2013) examine the safe haven characteristics of the Swiss franc. Habib and Stracca (2012) find that the net foreign asset position is a robust indicator of safe haven status. In a result analogous to the findings of this paper, Melvin and Taylor (2008) argue that conditioning the carry trade on a global financial stress index would have been a winning strategy during the Global Financial Crisis.

The existence of recurrent patterns in the cross-section of currency returns suggests that certain fundamental-based currency strategies could generate extraordinary profits, as explored by Nozaki (2010) and Jordà and Taylor (2012). Our finding of recurrent patterns in currency returns are consistent with the theoretical results by Gabaix and Maggiori (2014) which argue that changes in the risk bearing capacity of financiers may be behind risk-on/risk-off behavior of currencies.

## 2. Risk-off episodes

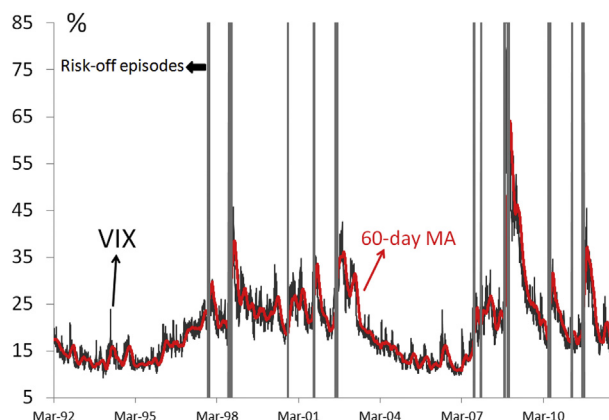
### 2.1. Identifying risk-off episodes

Our work requires an operational, quantitative, real-time definition of risk-off episodes. We experimented with a variety of rules based on widely tracked gauges of risk aversion in currency markets, such as the VIX, the AUDJPY exchange rate, measures of option-implied volatility on major currency pairs, and the cross-sectional volatility of exchange rates. In our view, the VIX presents three advantages. First, risk-off episodes derived from the VIX conform to our priors about disruptive market events in the last two decades. Second, the VIX is available at high frequencies, so the index can be used to identify the onset of risk-off episodes quickly. Third, the VIX is derived from options prices on the S&P 500 so does not directly draw on information from currency markets.

In our view, the start of a risk-off episode can be best defined using a non-linear threshold. For the objective of this paper, we chose to define the start of a risk-off episode when the VIX is 10 percentage points higher than its 60-day backward-looking moving average (MA). Under this definition, there

<sup>2</sup> In De Bock and de Carvalho Filho (2013) we show that deviations of actual exchange rates from measures of equilibrium levels also matter, as overvaluation in two of the three models of the IMF's Consultative Group on Exchange Rates (CGER) makes currencies more vulnerable in risk-off episodes.

<sup>3</sup> Safe haven currencies are defined as those that provide a hedge for a reference portfolio of risky assets, conditional on shocks to global risk aversion.



**Fig. 2.** VIX and risk-off episodes, 1992–2012.

Source: Bloomberg and authors' calculations

were 11 distinct risk-off episodes since 1992, as shown in Fig. 2. The rectangular shades in gray indicate periods of 12 weeks since the beginning of risk-off episodes. The spike in the VIX in fall 2008 dwarfs the other ones, but more recent episodes still indicate substantial increases in volatility relative to the baseline. It is also noteworthy that risk-off events have happened more often since 2007.

The risk-off episodes based on our rule are easily identifiable from their start date (Table 1). For instance, episode #1 is related to fears over the impact of the Asian crisis on the global economy and episode #2 can be traced back to concerns on Russia's economy that culminated into a devaluation and default on domestic bonds on 17 August 1998. Episode #3 is linked to wider macroeconomic concerns following the burst of the Dot Com bubble; #4 is the aftermath of the attack on the World Trade Center; and episode #5 is connected to fears of a slowing US economy. Episode #6 reflects increasing concerns on the valuation of subprime credit securities and asset-back commercial paper following the announcement by BNP Paribas that it could no longer value holdings of US subprime mortgage backed securities in some mutual funds. Episode #7 signals mounting disruptions in the USD money markets in the fall of 2007.<sup>4</sup> Episode #8 is the Lehman failure; #9 is the Greek crisis in May 2010; #10 is the aftermath of the Japan earthquake; and #11 is the bout of instability triggered by the confrontation over the US debt ceiling and a deterioration of the European crisis.<sup>5</sup>

In what follows, we focus on the 12 weeks subsequent to the beginning of a risk-off episode. We do not try to determine the end of a risk-off episode. Episodes of high volatility may last from a few weeks to several months, and the transition from turmoil to more accommodating financial conditions is usually harder to pinpoint than the onset of a risk-off period. The VIX shows some degree of mean reversion over longer horizons but for the risk-off episodes we study, there is little evidence of mean reversion. The average level of the VIX 12 weeks after the start of a risk-off episode is on average only 2.4 percentage points below the level that triggered the episode. This compares to a standard deviation of the change in the VIX of 9.8 percentage points for the same horizon of 12 weeks after the start of a risk-off episode.<sup>6</sup>

<sup>4</sup> We treat the two risk-off episodes in the second half of 2007 as a single risk-off episode, starting in August 2007. Both episodes are related to the U.S. subprime crisis and disruptions in the market for asset-backed commercial paper.

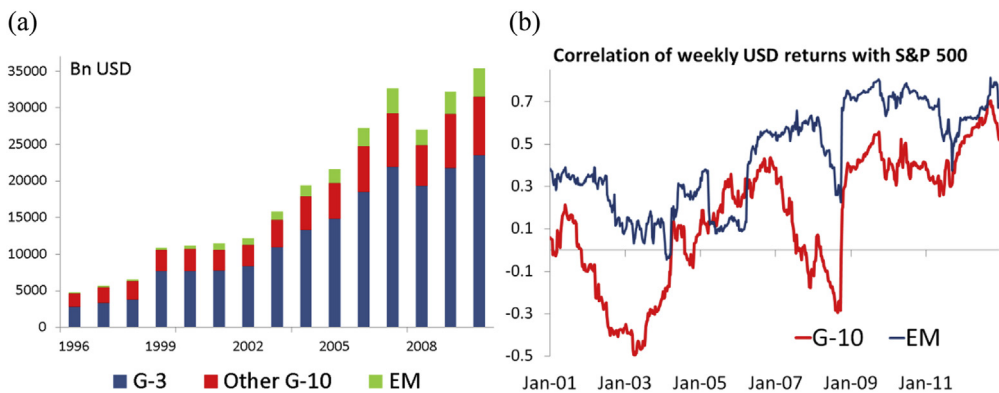
<sup>5</sup> The algorithm chooses September 17 2001 as the beginning of the 9/11 risk-off episode as US equity markets were closed for a few days and the VIX series was held at its 09/10 value until 9/16.

<sup>6</sup> This is calculated over the 8 episodes reported in Table 2.

**Table 1**

Initial dates of risk-off episodes.

| #  | Date              | Event                                                                       |
|----|-------------------|-----------------------------------------------------------------------------|
| 1  | 29 October 1997   | Escalation of Asian crisis                                                  |
| 2  | 4 August 1998     | Concerns on Russian economy                                                 |
| 3  | 12 October 2000   | Fear of slowing US economy                                                  |
| 4  | 17 September 2001 | 9/11 attacks                                                                |
| 5  | 10 July 2002      | Fear of slowing US economy                                                  |
| 6  | 10 August 2007    | BNP Paribas halts withdrawals from three money market mutual funds          |
| 7  | 12 November 2007  | Disruptions in USD money markets                                            |
| 8  | 17 September 2008 | Lehman failure                                                              |
| 9  | 6 May 2010        | Greek crisis                                                                |
| 10 | 16 March 2011     | Uncertainty over impact of Japan's March 11 earthquake                      |
| 11 | 4 August 2011     | Confrontation over US debt ceiling and deterioration of crisis in euro area |



**Fig. 3.** Increased financial integration and correlations. (a) Stock of portfolio liabilities. Note: Data from a country's international investment position. (b) FX returns and S&P 500. Note: Chart shows correlation of weekly returns on the S&P 500 with weekly USD returns on G-10 (based on the "DXY" index) and EM currencies ("FXJPEMCI"). Sample size is past 52 weeks. Source: Bloomberg, IFS and authors' calculations

## 2.2. Why have risk-off episodes become more frequent?

The initial dates of our risk-off episodes suggest that latent risks in the G-3 economies are driving the higher frequency of risk-off episodes. In fact, with the exception of the Japanese earthquake in March 2011, recent risk-off episodes can be traced back to concerns on sovereign or private sector balance sheets in Europe and the United States. Since 2007 investors are increasingly more worried about tail risks to global financial stability emanating from the euro area and the United States. Some observers argue that we have entered a world with multiple equilibria, where adequate policy responses could lead to a virtuous cycle and vice versa (e.g., IMF, 2011). Risk-off episodes bring tail risks to the forefront and put asset valuations in other developed and emerging markets under pressure.

In addition, the increased integration of global financial markets could lead to a more rapid repricing of global risks across markets. Taylor (2012) notes that the expansion of external balance sheets in the last two decades is unparalleled in economic history. The growth in cross-border financial flows reflects asset reallocation by Western investors, but also reserve accumulation, wealth creation and demographic developments in emerging economies. After some retrenchment in capital flows following Lehman's bankruptcy, the growth in international portfolio flows has resumed (Fig. 3).<sup>7</sup>

<sup>7</sup> Milesi-Ferretti and Tille (2011) discuss the factors behind the increase in home bias observed in 2008–09.

The larger stock of portfolio liabilities worldwide has important implications for currency markets. Sizable flows into local bond and equity markets have lead to a sharp increase in local currency exposure by foreign investors.<sup>8</sup> Even when there is no pullback of foreign flows in the wake of a risk-off event, hedging of local currency exposure can lead to FX weakness. In addition, periods of strong inflows tend to appreciate currencies, making undervaluations less evident. This might make investors less willing to stay long when a risk-off episode hits.

### 3. What happens to exchange rates during risk-off episodes?

In this section, we establish recurring features of currency movements during risk-off episodes.<sup>9</sup> We present some simple correlations before turning to a VAR analysis. We do not include the risk-off episodes of October 1997 and August 1998 as data are not available for some of the variables studied in the next section.

#### 3.1. Are risk-off episodes alike?

There are recurrent patterns in the cross-sectional distribution of risk-off depreciations. In general, there is a positive correlation between USD spot returns of different currencies across episodes (Table 2).<sup>10</sup> For a short horizon of three weeks, exchange rate behavior in previous episodes helps to predict risk-off depreciations in subsequent ones. For the horizon of twelve weeks, the patterns are harder to discern and no risk-off episodes stand out.

The risk-off episode related to the Japan earthquake in March 2011 is an exception to the rule and significantly negatively correlated with the other episodes. Looking closer at the episode explains why. Higher yielding currencies sold off right after the earthquake struck on March 11 but currency weakness was reversed after the coordinated currency intervention of March 18 (for details, see Neely, 2011). Although this might argue for excluding the Japan earthquake episode from our analysis, we choose to not let any ex-post knowledge guide our sample selection, so we include this episode.

#### 3.2. Evidence from local projections

We look at impulse response functions to measure the impact of a risk-off episode on a currency, we estimate impulse response functions using the method of local linear projections of Jordà (2005), using weekly data, from the first week of 2000 to June 2012.

In Jordà's formulation, an impulse response is defined as the difference between two forecasts:

$$IR(t, s, risk_t) \equiv E(\mathbf{Y}_{t+s} | risk_t = 1; \mathbf{X}_t) - E(\mathbf{Y}_{t+s} | risk_t = 0; \mathbf{X}_t) \quad \text{for } s = 1, 2, \dots, h \quad (1)$$

where the operator  $E(\cdot)$  denotes the best mean squared error predictor;  $\mathbf{Y}$  stands for the bilateral exchange rate with respect to the US dollar (measured in logs);  $risk_t$  denotes a dummy equal to one in the quarters when risk-off episodes started; and  $\mathbf{X}_t \equiv (\mathbf{Y}_{t-1}, \dots, \mathbf{Y}_{t-p}, risk_{t-1}, \dots, risk_{t-p})'$ .

This difference in forecasts can be estimated from the model below:

$$\mathbf{Y}_{t+s} = risk_t \beta_s + \alpha_s + \sum_{\tau=1..p} (\mathbf{Y}_{t-\tau} \mathbf{A}_\tau^s + risk_{t-\tau} \beta_\tau^s) + \mathbf{v}_{t+s}^s \quad \text{for } s = 1, 2, \dots, h \quad (2)$$

<sup>8</sup> Note that investors tend to hedge FX exposure related to fixed income versus equity holdings differently, and the composition of portfolio flows since 2008 may have had implications for currency markets as well.

<sup>9</sup> Previous research has found that in the aftermath of Lehman's failure, currencies whose returns were more correlated to the AUD/JPY, had higher interest rates, and were in an appreciation streak before the crisis tended to depreciate relatively to the U.S. dollar (e.g. De Bock and Englander, 2008). Currencies from countries running large current account deficits also depreciated more than others.

<sup>10</sup> In this paper, we focus on spot returns of local currencies against the USD during a risk-off episode. But it is worth noting that many FX transactions are against the euro in the case of some Central and Eastern European economies, whereas some countries have also pegged exchange rates at times. For instance, the Moroccan dirham is pegged to a disclosed basket with euro representing 80 percent and USD 20 percent.

**Table 2**

Correlation of spot returns across episodes.

| Horizon of 3 weeks  | October 2000 | September 2001 | July 2002 | August 2007 | Lehman Failure | Greece May 2010 | Japan March 2011 | August 2011 |
|---------------------|--------------|----------------|-----------|-------------|----------------|-----------------|------------------|-------------|
| October 2000        | 1.00         | 0.22           | 0.05      | 0.18        | 0.06           | 0.12            | 0.00             | 0.09        |
| September 2001      |              | 1.00           | 0.21      | 0.36        | 0.31           | 0.16            | -0.35            | 0.53        |
| July 2002           |              |                | 1.00      | 0.31        | 0.34           | 0.03            | -0.19            | 0.11        |
| August 2007         |              |                |           | 1.00        | 0.30           | 0.27            | -0.31            | 0.25        |
| Lehman Failure      |              |                |           |             | 1.00           | 0.47            | -0.70            | 0.42        |
| Greece May 2010     |              |                |           |             |                | 1.00            | -0.64            | 0.18        |
| Japan March 2011    |              |                |           |             |                |                 | 1.00             | -0.36       |
| August 2011         |              |                |           |             |                |                 |                  | 1.00        |
| Horizon of 12 weeks | October 2000 | September 2001 | July 2002 | August 2007 | Lehman Failure | Greece May 2010 | Japan March 2011 | August 2011 |
| October 2000        | 1.00         | 0.13           | 0.21      | 0.45        | -0.24          | 0.43            | 0.53             | -0.13       |
| September 2001      |              | 1.00           | -0.18     | -0.03       | 0.00           | -0.08           | 0.07             | 0.09        |
| July 2002           |              |                | 1.00      | -0.16       | 0.16           | -0.16           | -0.16            | 0.22        |
| August 2007         |              |                |           | 1.00        | -0.34          | 0.20            | 0.28             | -0.21       |
| Lehman Failure      |              |                |           |             | 1.00           | 0.05            | -0.31            | 0.42        |
| Greece May 2010     |              |                |           |             |                | 1.00            | 0.26             | -0.02       |
| Japan March 2011    |              |                |           |             |                |                 | 1.00             | -0.51       |
| August 2011         |              |                |           |             |                |                 |                  | 1.00        |

Note: Matrices show correlations between USD spot returns at three and twelve weeks into different episodes. Blue/red shaded cells denote positive/negative statistically significant correlations at the 95 percent level.

where the  $\mathbf{A}_\tau^{s+1}$  and  $\beta_\tau^{s+1}$  are matrices of coefficients describing the dynamics of the endogenous variables, including the effect of risk-off episodes prior to  $t$ ;  $\beta_s$  measures the impact of a risk-off episode occurring in time  $t$  on the best mean-squared minimizing forecast of  $\mathbf{Y}$  in periods  $t$  through  $t+h$ ; and the errors  $\mathbf{v}_{t+s}$  have mean zero but are serially correlated up to  $sth$ -order. For each forecast horizon, the method allows for a flexible choice of the lag length  $p$ , based on the optimization of the Akaike Information Criterion, constrained to not more than 4 lags. The impulse response function is then estimated by  $IR(t, s, risk_t) = \hat{\beta}_s$  and its confidence intervals are based on the standard error for the parameter  $\beta_s$ , using a Newey type estimator allowing for heteroskedasticity and serial correlation in the  $\mathbf{v}_{t+s}$  residuals.

In the estimation of the impulse response functions to a risk-off episode, we consider for the endogenous variable ( $\mathbf{Y}$ ) the weekly closing value of the bilateral nominal exchange rate relative to the U.S. dollar (from the IMF/GDS database). We also include a linear and a quadratic trend as control variables to capture longer-term structural changes (e.g. some countries have reduced inflation rates during the sample period).

We find significant immediate and cumulative effects of risk-off episodes on currency performance (Table 3). The Japanese yen and the Swiss franc appreciate relatively to the U.S. dollar at the beginning of a risk-off episode.<sup>11</sup> The appreciation of the Swiss franc tends to be reverted starting the third week of the risk-off episode, while the appreciation of the Japanese yen is more persistent. After 12 weeks, the cumulative risk-off appreciation of the Japanese yen is 3.7 percent.

On the other side of the spectrum, the Australian and New Zealand dollars, Brazilian real, Chilean, Colombian and Mexican peso, Indian rupee, Malaysian ringgit, Peruvian sol, South African rand, Thai baht and Turkish lira, depreciate either on impact or in the week immediately after the risk-off episode

<sup>11</sup> On a multilateral basis, the U.S. dollar also appreciates during risk-off episodes, so our convention of measuring all exchange rates relative to the U.S. dollar tends to somewhat understate the extent to which the Japanese yen and Swiss franc exhibit safe haven behavior.

**Table 3**

Impulse response functions, effect of risk-off episodes on exchange rates versus the U.S. dollar.

|                    | Number of weeks since the beginning of the risk-off episode |              |              |              |              |
|--------------------|-------------------------------------------------------------|--------------|--------------|--------------|--------------|
|                    | 0                                                           | 1            | 3            | 6            | 12           |
| Australia          | 1.73 [0.80]                                                 | 3.31 [1.30]  | 6.49 [3.17]  | 4.67 [3.09]  | 3.56 [3.20]  |
| Brazil             | 2.77 [0.94]                                                 | 3.04 [1.13]  | 7.10 [2.84]  | 4.95 [2.45]  | 6.02 [5.24]  |
| Canada             | 0.94 [0.58]                                                 | 0.91 [0.65]  | 3.15 [1.56]  | 2.11 [2.09]  | 2.81 [2.87]  |
| Chile              | 0.69 [0.26]                                                 | 1.15 [0.56]  | 2.68 [1.82]  | 3.56 [3.13]  | 4.21 [3.16]  |
| Colombia           | 1.93 [0.59]                                                 | 2.34 [1.10]  | 3.86 [1.82]  | 2.98 [2.31]  | 3.50 [2.88]  |
| Czech Republic     | -0.01 [1.01]                                                | -0.46 [1.10] | 1.06 [1.73]  | 2.20 [2.35]  | 0.37 [2.49]  |
| Euro Area          | -0.23 [0.78]                                                | 0.23 [1.29]  | 1.05 [1.51]  | 2.21 [2.04]  | 0.46 [1.32]  |
| Hungary            | 1.02 [1.52]                                                 | 0.22 [2.01]  | 1.64 [2.61]  | 3.43 [3.57]  | 2.02 [3.10]  |
| India              | 0.88 [0.34]                                                 | 1.12 [0.55]  | 1.94 [1.07]  | 1.84 [1.33]  | 1.64 [1.47]  |
| Indonesia          | 0.84 [0.57]                                                 | 0.55 [0.50]  | 2.34 [0.92]  | 3.79 [2.30]  | 3.75 [2.43]  |
| Israel             | 0.02 [0.66]                                                 | -0.72 [1.04] | -0.20 [1.25] | -0.14 [1.68] | -0.09 [1.95] |
| Japan              | -1.09 [0.62]                                                | -2.54 [0.66] | -1.56 [1.13] | -1.94 [1.02] | -3.71 [2.26] |
| Korea              | 0.83 [0.61]                                                 | 1.45 [0.78]  | 5.35 [3.34]  | 3.50 [2.29]  | 3.91 [2.75]  |
| Malaysia           | 0.78 [0.39]                                                 | 0.31 [0.31]  | 1.13 [0.54]  | 1.22 [0.83]  | 0.79 [0.85]  |
| Mexico             | 1.19 [0.77]                                                 | 1.33 [0.73]  | 4.51 [2.78]  | 3.19 [2.72]  | 4.18 [3.12]  |
| New Zealand        | 1.50 [0.88]                                                 | 2.30 [1.85]  | 4.59 [1.67]  | 3.48 [2.04]  | 3.34 [2.84]  |
| Norway             | 0.55 [0.94]                                                 | -0.02 [1.09] | 1.62 [1.65]  | 2.38 [2.61]  | 1.73 [2.76]  |
| Peru               | 0.28 [0.09]                                                 | 0.08 [0.22]  | 0.92 [0.42]  | 0.95 [0.58]  | 0.43 [0.99]  |
| Philippines        | 0.17 [0.37]                                                 | 0.21 [0.33]  | 1.28 [0.58]  | 1.75 [0.72]  | 1.09 [0.86]  |
| Poland             | 1.46 [1.56]                                                 | 0.50 [1.58]  | 2.50 [1.98]  | 3.55 [3.50]  | 1.78 [3.78]  |
| Romania            | 0.50 [0.98]                                                 | 0.69 [1.64]  | 1.35 [1.95]  | 2.27 [2.70]  | 1.29 [2.27]  |
| Russian Federation | 0.77 [0.70]                                                 | 0.69 [0.99]  | 1.15 [0.74]  | 1.94 [1.67]  | 1.74 [1.60]  |
| Singapore          | 0.34 [0.43]                                                 | -0.16 [0.46] | 1.08 [0.62]  | 1.26 [0.94]  | 1.22 [1.16]  |
| South Africa       | 1.31 [0.75]                                                 | 1.77 [0.99]  | 4.20 [1.99]  | 6.12 [3.04]  | 7.25 [4.43]  |
| Sweden             | 1.22 [0.99]                                                 | 1.31 [1.02]  | 2.18 [1.29]  | 2.41 [2.33]  | 1.26 [2.48]  |
| Switzerland        | -1.63 [0.82]                                                | -2.37 [1.35] | -0.45 [1.20] | 0.78 [1.79]  | -0.42 [2.04] |
| Thailand           | -0.54 [0.31]                                                | -0.49 [0.61] | 0.36 [0.38]  | 0.85 [0.47]  | 1.25 [0.81]  |
| Turkey             | 1.76 [0.67]                                                 | 2.86 [1.47]  | 4.41 [1.83]  | 3.83 [2.91]  | 0.99 [3.39]  |
| United Kingdom     | -0.28 [0.63]                                                | 0.36 [1.02]  | 0.68 [1.10]  | 2.28 [1.51]  | 1.68 [2.49]  |

The impulse response functions were derived using the method of local linear projections (Jordà, 2005), as specified in equation (2), controlling for linear, quadratic and cubic trends. The sample covers January 2000 to June 8 2012. Standard errors are recorded inside square brackets, and allow for heteroskedasticity and serial correlation. Numbers with a color background are statistically significant at the 10 percent level (two-sided test) and green shades denote appreciation in a risk-off episode while orange background denotes risk-off depreciation.

starts. One week after the start of the risk-off episode, the Australian dollar and the Brazilian real depreciate on average by more than 3 percent.

Other currencies tend to depreciate three to six weeks into the risk-off episode, namely the Canadian and Singaporean dollars, Indonesian rupiah, Polish zloty, Swedish krona and Turkish lira, are found to depreciate over a horizon of 3–6 weeks into the risk-off episode. The inclusion of the Swedish krona in this sample is remarkable given its strong appreciation versus the euro in May–July 2012, when euro area concerns flared up. Market participants note that the krona has traded more like a safe haven during these months. After 12 weeks, the cumulative risk-off depreciations of the Brazilian real, Chilean peso, South African rand and Mexican peso are more than 4 percent.



While some currencies do not show a statistically significant response to risk-off episodes here, it is worth pointing out that we lump risk-off episodes together in the VAR. The lack of significance may reflect wide changes in a currency's sensitivity to a risk-off conditions over time (that is, across the episodes we study). For example, the Russian ruble weakened substantially in the fall of 2008 but did not underperform when we pool episodes since 2000. We could also not find any significant effect for the currencies of the Czech Republic, euro area, Hungary, Israel, Norway, Romania and United Kingdom.

Interestingly, there is quite a bit of heterogeneity among the currencies that depreciate in a risk-off episode. The impulse response function for the currencies of advanced economies, such as Australia, New Zealand and Sweden but also Turkey show mean reversion as weeks go by. On the other hand, weakness of EM currencies such as Brazil and South Africa is more persistent, as the risk-off depreciations show little tendency towards mean reversion, at least in the quarter following the on-set of the risk-off event.

#### 4. Explaining the cross-sectional variation

In this section we show that a number of factors are associated with FX performance during risk-off episodes. We want to emphasize that all our variables are available *prior* to the risk-off episode and we do not include later data points in the analysis. Interestingly, our results suggest that investors purposefully re-evaluate currency risk during risk-off episodes without necessarily updating their information set.

Economic theory and previous work dictate our choice of explanatory variables, while we also add variables on option prices and liquidity. Though some of the variables are highly correlated, it is worth to consider the cross-sectional patterns of depreciation associated to each variable in isolation before we move to multivariate regressions.

##### 4.1. Policy interest rates

- *Policy interest rate before the episode.* High interest rate currencies tend to be on the receiving end of carry trade and portfolio flows. Market conditions that cause a pullback or slowdown in these flows would weaken high interest rate currencies.<sup>12</sup> It is worth noting that policy interest rates are often set in response to developments in the domestic economy. As such, policy rates are highly correlated with inflation rates.<sup>13</sup>

##### 4.2. External sector

- *Lagged current account balance.* The availability of financing of current account deficits may become more restrictive in risk-off episodes, so we expect larger depreciations in countries with larger deficits during such episodes. [Fratzscher \(2009\)](#) finds that during the global financial crisis, the currencies of countries with larger current account deficits experienced substantially larger depreciations.
- *Net foreign asset position.* A country with a large positive net foreign asset position (or reserves) is less likely to run into problems servicing its external debt or face liquidity shortages that lead to currency sell-offs.<sup>14</sup> In particular, a larger stock of reserves relative to short-term liabilities can help to defend the exchange rate level during risk-off episodes. [McCauley \(2012\)](#) shows that many EM sold substantial amounts of foreign exchange reserves and U.S. dollars forward in 2008 to defend their currencies. In addition, repatriation flows could very well strengthen the currency. This seems to be the case in Switzerland ([de Carvalho Filho, 2015](#)).

<sup>12</sup> Policy interest rates are from IMF/GDS, extended by the IFS series 'short rate' for Indonesia, Malaysia and Mexico.

<sup>13</sup> Inflation and policy rates at the onset of a risk-off episode are highly correlated (80%) in the cross-section. If purchasing power parity holds, a higher rate of inflation would also imply a higher rate of depreciation.

<sup>14</sup> [Habib and Stracca \(2012\)](#) find that the net foreign asset position is a robust determinant of safe haven currency status.

- *Financial openness.* Openness is measured as the sum of the absolute value of portfolio inflows and outflows the year before the risk-off episode (as a ratio to GDP). With this variable we want to capture potential vulnerability to currency weakness following a period of substantial flow activity as investor decide to reallocate in a risk-off episode. In addition, liquidity conditions generally declined in secondary markets in recent years whereas nonresident holdings of local currency bonds increased ([International Monetary Fund, 2013](#)). Currency fluctuations could then be large if market makers such as local or international banks struggle to absorb outflows. Even without outflows, hedging activities of foreign investors can also lead to large exchange rate moves in the spot market in risk-off episode.

#### 4.3. Currency behavior prior to the risk-off episode

- *Beta with respect to AUDJPY.* Many FX investors track the highly liquid Japanese yen per Australian dollar (AUDJPY) to evaluate broad risk positioning on FX markets. To get an indication of how investors position a currency across the currency universe, we calculate the beta of a currency with the AUDJPY in the 52 weeks prior to the episode.<sup>15</sup> A positive beta means that a currency tends to depreciate when AUDJPY is weaker.

#### 4.4. Cost of buying an option and tail risk insurance

- *Implied volatility of at-the-money options with a 3-month maturity.* This implied volatility is a forward-looking measure of the cost investors are willing to pay to hedge against USD moves from current market levels over the next 3 months.<sup>16</sup> Measures of implied volatility also tend to be correlated with realized volatility.
- *Risk reversals versus U.S. dollar.* Risk reversals track the premium investors are willing to pay for insuring against sharp currency swings. This insurance cost gives an indication of market concerns on tail risk scenarios. To calculate the risk reversals of a given currency, we subtract the implied volatility of an out-of-the-money (25 delta) USD put option at the 3-month maturity from the implied volatility of an equivalent out-of-the-money USD call option.<sup>17</sup> In this case, a *higher* value of a risk reversal implies that investors are willing to pay *more* to insure against weakness vis-à-vis the USD in a risk-off episode. We expect risk reversals to be positively correlated with currency weakness during a risk-off episode.

#### 4.5. Liquidity conditions and benchmark dynamics

- *Bid-ask spread.* The bid-ask spread captures the ease at which investors can unwind or build positions in a currency. The effect of bid-ask spreads on a currency's exposure to risk-off episodes is an empirical question. It is plausible that currencies that are more efficiently traded are the first ones that investors would sell off during a risk-off episode. But low bid-ask spreads may also indicate deeper markets. This could allow investors to unwind positions with more muted effects on prices. The measure we use is based on the median bid-ask spread in the month before the episode, from Datastream.
- *Role of benchmark investors.* A country's exposure to global risk aversion shocks can also depend on its importance to benchmark investors. This is particularly relevant for emerging markets, with many bond funds following dedicated mandates and focusing more heavily on countries with large shares in benchmark indices. We will track this variable as the share of each country's bond market in the universe of emerging market bond markets (for example: Brazil's share is about 15%; Chile's share is 0.5%), using EMTA data.

<sup>15</sup> De Bock and Englander (2008) find that this beta was significantly correlated with FX performance in 2008.

<sup>16</sup> The implied volatility is effectively the market price of an option when we know the strike of the option, the current spot/forward level of an exchange rate, interest rates and time to maturity.

<sup>17</sup> The "smile" refers to the observation that implied volatilities of out-of-the-money calls and puts are different from options struck at-the-money.

**Table 4**

Simple determinants of exchange rate response to risk-off episodes. We estimate the equation:  $\mathbf{Y}_{it+s} = \lambda_t + X_{it-1}(\alpha_s + \beta_s \text{risk}_t) + \dots + \sum_{\tau=1..p} (\mathbf{Y}_{t-\tau} \mathbf{A}_\tau^{s+1} + X_{it-\tau-1}(\alpha_\tau^{s+1} + \beta_\tau^{s+1} \text{risk}_{t-\tau})) + \mathbf{v}_{it+s}^s$ . The coefficients reported in the table are the  $\beta$  for each dependent variable  $X$ , time horizon  $s$ .

| Dependent variable: Depreciation since the beginning of the risk-off                           |                     | Number of weeks since the episode started |                     |                    |                     |
|------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------|---------------------|--------------------|---------------------|
|                                                                                                | Mean<br>[St. Dev]   | 1                                         | 3                   | 6                  | 12                  |
| (1) Emerging market dummy                                                                      | 0.600<br>[0.49]     | 0.4518<br>[0.238]                         | -0.0446<br>[0.3381] | 0.1593<br>[0.451]  | 0.6420<br>[0.6172]  |
| <b>Policy Rate</b>                                                                             |                     |                                           |                     |                    |                     |
| (2) Interest rate before the episode                                                           | 6.155<br>[7.01]     | 0.1475<br>[0.0206]                        | 0.1992<br>[0.0292]  | 0.1657<br>[0.039]  | 0.1063<br>[0.0533]  |
| <b>External sector</b>                                                                         |                     |                                           |                     |                    |                     |
| (3) Current account balance (% GDP)                                                            | 1.568<br>[6.622]    | -0.087<br>[0.0173]                        | -0.112<br>[0.0246]  | -0.107<br>[0.0328] | -0.114<br>[0.0449]  |
| (4) Net foreign assets (% GDP)                                                                 | -8.275<br>[62.412]  | -0.011<br>[0.0018]                        | -0.012<br>[0.0026]  | -0.011<br>[0.0034] | -0.012<br>[0.0047]  |
| (5) Financial openness<br>(sum of the absolute value of portfolio inflows and outflows to GDP) | -0.025<br>[0.016]   | -0.020<br>[0.013]                         | -0.0253<br>[0.0184] | 0.0156<br>[0.0246] | 0.0219<br>[0.0337]  |
| <b>Sensitivity to global and FX-specific risks</b>                                             |                     |                                           |                     |                    |                     |
| (6) Beta with AUD/JPY exchange rate                                                            | 0.152<br>[0.222]    | 5.0284<br>[0.6025]                        | 6.3035<br>[0.855]   | 5.3041<br>[1.1398] | 7.2412<br>[1.5525]  |
| <b>Option prices</b>                                                                           |                     |                                           |                     |                    |                     |
| (7) Implied volatility of at-the-money option                                                  | 11.920<br>[5.606]   | 0.1386<br>[0.0276]                        | 0.2122<br>[0.0393]  | 0.2272<br>[0.0523] | 0.2268<br>[0.0713]  |
| (8) Risk reversal versus USD (tail risk insurance)                                             | 1.290<br>[2.186]    | 0.6534<br>[0.0835]                        | 0.8392<br>[0.1165]  | 0.7165<br>[0.1525] | 1.0530<br>[0.2038]  |
| <b>Liquidity measures and financial openness</b>                                               |                     |                                           |                     |                    |                     |
| (9) Bid-ask spread, in bps                                                                     | 12.960<br>[120.797] | 0.0103<br>[0.0052]                        | 0.0184<br>[0.0074]  | 0.0221<br>[0.0099] | 0.0215<br>[0.0136]  |
| (10) Market size<br>(country's share of total EM trading volume; source: EMTA)                 | 0.038<br>[0.064]    | 7.6706<br>[2.6094]                        | 17.5915<br>[3.7471] | 8.5594<br>[5.0542] | 18.1605<br>[6.9635] |

Note: This table displays estimates of the slope of the impulse response function as in equation (4) with respect to each one of the country characteristics. Standard errors are recorded inside square brackets, and allow for heteroskedasticity and serial correlation. Shaded numbers are statistically significant at the 95% level (two-sided test). Green shade denotes variables associated with an appreciation during a risk-off episode, orange background denotes a risk-off depreciation. The sample includes Australia, Brazil, Canada, Chile, Colombia, Czech Republic, Euro Area, Hungary, India, Indonesia, Israel, Japan, Republic of Korea, Malaysia, Mexico, New Zealand, Norway, Peru, Philippines, Poland, Romania, Russian Federation, Singapore, South Africa, Sweden, Switzerland, Taiwan, Thailand, Turkey, and United Kingdom.

#### 4.6. Simple regressions

The rows of Table 4 present the Jordà-style impulse response functions describing the response of depreciation against the USD to a risk-off episode, where we interact the risk-off variable with the country-specific variables  $X_i$  described above.

Algebraically, we estimate:

$$\mathbf{Y}_{it+s} = \lambda_t + X_{it-1}(\alpha_s + \beta_s \text{risk}_t) + \dots + \sum_{\tau=1..p} (\mathbf{Y}_{t-\tau} \mathbf{A}_\tau^s + X_{it-\tau-1}(\alpha_\tau^s + \beta_\tau^s \text{risk}_{t-\tau})) + \mathbf{v}_{it+s}^s \quad \text{for } s = 1, 2, \dots, h \quad (3)$$

Thus we obtain impulse response functions that depend on country characteristics.

$$IR(t, s, \text{risk}_t | X_{it-1}) = \kappa_s + \hat{\beta}_s X_{it-1} \quad \text{for } s = 0, 1, 2, \dots, h \quad (4)$$

We focus on four different horizons  $k$  (one, 3-, 6- and 12-weeks), and estimate equation (3) on the same sample as the exercise of the previous section. The sample of currencies is determined by data availability and includes advanced and emerging markets.

First, EM currencies (row 1 in Table 4) do not differ statistically from advanced economies from the onset of the risk-off episode to horizons of up to twelve weeks. However, the evidence from the point estimates are suggestive that after a risk-off episode, EM currencies depreciate more over longer horizons. This result is consistent with the impulse response functions in section 3. Currencies of advanced economies such as the Australian and New Zealand dollars tend to depreciate on impact but also rebound sharply. In contrast, EM currencies such as the Brazilian real or the Mexican peso face more persistent depreciations on impact.

We find that high-yielding currencies (row 2) tend to depreciate more after a risk-off episode, with their peak depreciation around the third week following the onset of a risk-off episode. That is likely related to disruptions in carry trade positioning by global investors.

Balance of payments determinants are also correlated with risk-off currency movements. Higher current account balances (3) and stronger net foreign asset positions (4) are correlated with appreciation over all horizons.

On the other hand, simple regressions suggest that countries that were more financially open *de facto* (5) in the run-up to a risk-off episode are not more vulnerable to FX weakness.

Sensitivity to global trading trends, measured by the beta with respect to the AUDJPY, has a strong unconditional correlation with risk-off depreciations at all horizons (6). It is worth repeating that we calculate the betas of currency returns with the AUDJPY *prior* to the risk-off episode.

Option prices also turn up significant in the regressions. Both implied volatility (7) and risk reversals (8) are positively correlated with risk-off depreciations. This is reassuring, as these variables track how costly it is for investors to insure against currency moves from current levels. We would expect dealers to charge higher prices for options on currencies that are more sensitive to sell-offs during a risk-off episode. In contrast, it is worth pointing out that option prices are not a good predictor of currency moves in tranquil times and some analysts argue that implied volatilities underprice tail risks.<sup>18</sup>

Market-liquidity factors matter as well. Bid-ask spreads (9) are associated with risk-off depreciations – currencies with larger bid-ask spreads tend to depreciate more on impact and up to 6 weeks. In addition, currencies of countries whose bonds are actively traded also tend to sell off in a risk-off episode (10).

#### 4.7. Multivariate analysis

Next we distill the essential variables from Table 4 and use these in multivariate regressions. We focus on two different horizons (one- and 12-week), and estimate a multivariate version of equation (4) in the previous subsection. We split the factors into those representing macroeconomic/balance of payments variables and those representing market prices. This distinction is relevant because market prices tend to aggregate macroeconomic and other information and inclusion of market prices may drive down the economic and statistical significance of macroeconomic factors and give the erroneous impression that those do not matter.

We start with a specification with eight explanatory variables, which we chose based on the fit of the regressions estimated above while excluding collinear variables (Table 5).<sup>19</sup> For each horizon we present three specifications (i) a regression with macroeconomic variables and market prices, (ii) only macroeconomic variables, and (iii) only market price variables. Each specification also includes the

<sup>18</sup> For example, Englander (2012) argues that EURUSD risk reversals can look relatively cheap compared to indicators for euro area sovereign risks.

<sup>19</sup> We also used the R package BMS (Zeugner, 2010) to estimate Bayesian model averaging of linear specifications for the effect of country characteristics on depreciation since a risk-off episode with all the variables in Table 4 and a range of other variables. We included balance of payments and international investment position variables as well as financial development measures from Beck and Demirgüç-Kunt (2009), such as financial system deposits (% GDP) and stock market turnover ratio. The sign of the posterior means of the coefficients are in line with those reported in Table 5. One week into the episode, the policy interest rate has a large posterior inclusion probability (PIP). In week three, the currency beta with the AUDJPY has the largest PIPs. In the twelfth week, currency betas still have the highest PIPs, and the emerging market dummy, the option-implied volatility before the episode, and the trading volume of the domestic bond market have PIPs over 1/3. Those results are available upon request.

**Table 5**  
Multivariate determinants of exchange rate response to risk-off episodes. We estimate the equation:  
$$\mathbf{Y}_{it+s} = \lambda_t + \mathbf{X}_{it-1}(\alpha_s + \beta_s risk_t) + \dots + \sum_{\tau=1,p} (\mathbf{Y}_{t-\tau} \mathbf{A}_\tau^{s+1} + \mathbf{X}_{it-\tau-1}(\alpha_\tau^{s+1} + \beta_\tau^{s+1} risk_{t-\tau})) + \mathbf{v}_{it+s}^s$$
  
The coefficients reported in the table are the  $\beta_s$  for the vector of country characteristics  $\mathbf{X}$ , and time horizon  $s$ .

| Dependent variable: Depreciation since the beginning of the risk-off episode               | Number of weeks since the episode started |                     |                     |                     |                     |                     |
|--------------------------------------------------------------------------------------------|-------------------------------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                                                                                            | 1                                         |                     | 12                  |                     | 12                  |                     |
| Emerging market dummy                                                                      | 0.0100<br>[0.3656]                        | -0.6554<br>[0.3297] | 0.0671<br>[0.2942]  | -0.9239<br>[0.89]   | -0.8155<br>[0.8025] | -0.0751<br>[0.7187] |
| <b>Macroeconomic variables</b>                                                             |                                           |                     |                     |                     |                     |                     |
| Interest rate before the episode                                                           | 0.0770<br>[0.0671]                        | 0.2707<br>[0.0561]  |                     | 0.4143<br>[0.1634]  | 0.5682<br>[0.1364]  |                     |
| Current account balance (% GDP)                                                            | -0.0241<br>[0.033]                        | -0.0374<br>[0.0314] |                     | -0.0292<br>[0.0802] | -0.0546<br>[0.0765] |                     |
| Net foreign assets (% GDP)                                                                 | -0.0009<br>[0.0032]                       | -0.0053<br>[0.0031] |                     | -0.0018<br>[0.0079] | -0.0060<br>[0.0076] |                     |
| Financial openness<br>(sum of the absolute value of portfolio inflows and outflows to GDP) | 0.0399<br>[0.0186]                        | 0.0579<br>[0.0175]  |                     | 0.0825<br>[0.0454]  | 0.1067<br>[0.0427]  |                     |
| <b>Market prices</b>                                                                       |                                           |                     |                     |                     |                     |                     |
| Beta with the AUDJPY exchange rate                                                         | 1.3714<br>[1.0773]                        |                     | 2.1782<br>[1.005]   | 2.2493<br>[2.6227]  |                     | 3.8624<br>[2.4548]  |
| Risk reversal                                                                              | 0.3963<br>[0.1341]                        |                     | 0.4205<br>[0.1226]  | 0.2982<br>[0.3265]  |                     | 0.5672<br>[0.2995]  |
| Implied volatility                                                                         | 0.0796<br>[0.0512]                        |                     | 0.0897<br>[0.046]   | -0.0131<br>[0.1246] |                     | 0.0435<br>[0.1123]  |
| Bid-ask spread, in bps                                                                     | -0.0194<br>[0.01]                         |                     | -0.0199<br>[0.0099] | 0.0412<br>[0.0243]  |                     | 0.0349<br>[0.0242]  |
| Number of observations                                                                     | 14143                                     | 14143               | 14143               | 13835               | 13835               | 13835               |
| R-squared                                                                                  | 0.461                                     | 0.459               | 0.461               | 0.549               | 0.548               | 0.546               |
| Adjusted R-squared                                                                         | 0.433                                     | 0.431               | 0.433               | 0.526               | 0.524               | 0.522               |
| RMSE                                                                                       | 1.701                                     | 1.704               | 1.701               | 4.141               | 4.147               | 4.155               |

Note: Standard errors are recorded inside square brackets, and allow for heteroskedasticity and serial correlation. Numbers in green shade are statistically significant at the 95 percent level (two-sided test) and denote risk-off appreciation, orange background denotes risk-off depreciation. The sample coverage includes Australia, Brazil, Canada, Chile, Colombia, Czech Republic, Euro Area, Hungary, India, Indonesia, Israel, Japan, Republic of Korea, Malaysia, Mexico, New Zealand, Norway, Peru, Philippines, Poland, Romania, Russian Federation, Singapore, South Africa, Sweden, Switzerland, Thailand, Turkey and United Kingdom.

emerging market dummy and the episode-specific dummies. The EM dummy turns up significant in 2 specifications, with a negative signal (controlling for other characteristics, emerging market depreciate less in response to a risk-off episode).

For the immediate effect of the risk-off episode (1-week horizon), financial openness and risks reversals are significant in the specification with all variables. Countries which have received and sent out larger financial flows tend to depreciate more in the week after the start of a risk-off episode. In the specification with only macroeconomic variables, high-yield currencies turn out to be more sensitive whereas risk reversals and the beta with the AUDJPY currency pair are significant in the specification with market prices.

Turning to the 12-week horizon, we see an increasing explanatory power of the empirical specifications (the adjusted R-squares rise from about 0.46 to 0.55). Interest rates are significant in the specification with all variables and the model with just macroeconomic variables.

## 5. Have currency risk factors changed since 2007?

This section examines if currency risk factors have changed in response to the evolving global economic environment. It is plausible that low interest rates in core advanced economies and recurring fears about global financial stability have changed the behavior of currencies during risk-off episodes since 2007. We again estimate impulse response functions to risk-off episodes using local linear projections (Jordà, 2005), using a specification where the response to a risk-off episode depends on country characteristics, but the impact of those country characteristics is allowed to vary after 2007. We estimate two specifications, one with currency risk factors representing macroeconomic and balance of payments variables and a second regression with factors more closely associated with market pricing.

We then compare the impact of macroeconomic variables and market prices before and after 2007 by introducing interaction terms for each one of the variables with a dummy for the episodes since 2007. The pre-2007 period includes 3 risk-off episodes (October 2000, September 2001 and July 2002), and the post-2007 period includes 5 risk-off episodes (the August 2007 subprime crisis, Lehman's bankruptcy, the Greek crisis in May 2010, Japan's March 2011 earthquake and the confrontation over the U.S. debt ceiling), for a total of 8 risk-off episodes.

A key finding is that the effect of the interest rate has become substantially larger after 2007 (Table 6, panel A). Since the global financial crisis, high-yield currencies have depreciated more sharply during risk-off episodes. The coefficients on the interest rate variable are also significant at every horizon. More over, the effect is larger the longer the horizon considered.

Since the global financial crisis, cuts in G-3 policy rates have prompted many investors to increase their exposure to high-yield currencies. The increased importance of the policy rate in recent risk-off episodes could reflect investors' decisions to hedge or scale back positions when the VIX spikes. In addition, there were widespread losses on leveraged carry trades in high-yielding currencies in 2008, and investors are probably tracking more closely the extent to which a currency trades as a carry play.<sup>20</sup>

While the effect of some macroeconomics characteristics on risk-off depreciations has changed since 2007, in general we could not find changes in the way market prices interact affect risk-off depreciations. The only exception were bid-ask spreads (Table 6, panel B). The beta with the AUDJPY exchange rate is significant in the pre-2007 sample but the interaction term with the dummy for episodes since 2007 is not. It is nevertheless worth noting that the median beta with the AUDJPY among emerging markets has trended up since 2007 (Fig. 4).

## 6. Conclusion and policy implications

The cross-sectional behavior of exchange rates during risk-off episodes displays some recurring patterns. In this paper, we take a descriptive approach and document the behavior of exchange rates after spikes in the VIX, the option-implied volatility of equity prices, what we call risk-off episodes.

<sup>20</sup> Haldane (2011) argues that the financial crash of 2008 has left risk-takers psychologically scarred and has made them more willing to use simple heuristics such as risk-on/risk-off. This could increase contagion in financial markets.

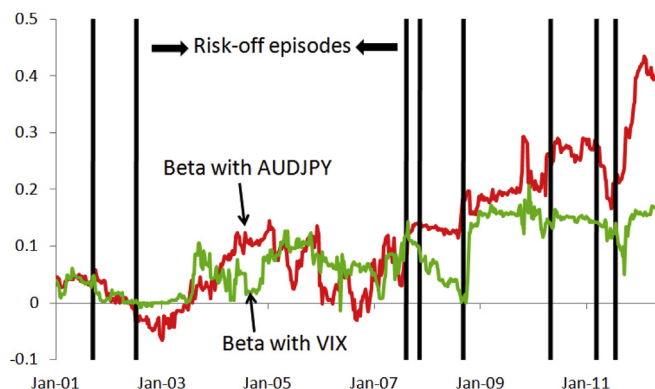
**Table 6**

Risk-off depreciations, before and after the global financial crisis.

Dependent variable: **Depreciation since the beginning of the risk-off episode**

| Panel A. Regression with macroeconomic variables  | Number of weeks since the episode started |                   |                   |                   |
|---------------------------------------------------|-------------------------------------------|-------------------|-------------------|-------------------|
|                                                   | 1                                         | 3                 | 6                 | 12                |
| Emerging market dummy                             | -0.030<br>[0.617]                         | 0.237<br>[0.872]  | -0.370<br>[1.159] | 2.534<br>[1.567]  |
| <i>X After 2007</i>                               | -0.706<br>[0.709]                         | -2.355<br>[1.003] | -0.506<br>[1.332] | -3.404<br>[1.801] |
| Interest rate before the episode                  | 0.091<br>[0.028]                          | 0.123<br>[0.04]   | 0.094<br>[0.053]  | -0.090<br>[0.072] |
| <i>X After 2007</i>                               | 0.214<br>[0.065]                          | 0.336<br>[0.092]  | 0.301<br>[0.123]  | 0.647<br>[0.166]  |
| Current account balance (% GDP)                   | -0.024<br>[0.072]                         | 0.085<br>[0.102]  | 0.029<br>[0.135]  | -0.085<br>[0.183] |
| <i>X After 2007</i>                               | -0.014<br>[0.079]                         | -0.145<br>[0.112] | -0.130<br>[0.149] | 0.000<br>[0.201]  |
| Net foreign assets (% GDP)                        | -0.018<br>[0.008]                         | -0.022<br>[0.011] | -0.014<br>[0.014] | -0.014<br>[0.019] |
| <i>X After 2007</i>                               | 0.014<br>[0.008]                          | 0.020<br>[0.012]  | 0.013<br>[0.016]  | 0.014<br>[0.021]  |
| Financial openness                                | 0.023<br>[0.038]                          | 0.053<br>[0.054]  | 0.088<br>[0.072]  | 0.208<br>[0.097]  |
| <i>X After 2007</i>                               | 0.029<br>[0.043]                          | -0.026<br>[0.06]  | 0.012<br>[0.08]   | -0.111<br>[0.108] |
| Number of observations                            | 15666                                     | 15604             | 15511             | 15340             |
| R-squared                                         | 0.410                                     | 0.441             | 0.457             | 0.480             |
| Adjusted R-squared                                | 0.385                                     | 0.419             | 0.434             | 0.458             |
| RMSE                                              | 1.807                                     | 2.554             | 3.394             | 4.588             |
| Panel B. Regression with market pricing variables | Number of weeks since the episode started |                   |                   |                   |
|                                                   | 1                                         | 3                 | 6                 | 12                |
| Emerging market dummy                             | 1.492<br>[0.576]                          | 1.393<br>[0.815]  | 0.193<br>[1.08]   | 1.913<br>[1.466]  |
| <i>X After 2007</i>                               | -1.151<br>[0.65]                          | -1.989<br>[0.919] | -0.339<br>[1.218] | -1.812<br>[1.654] |
| Beta with the AUDJPY exchange rate                | 4.573<br>[1.201]                          | 2.668<br>[1.699]  | 0.782<br>[2.252]  | 2.808<br>[3.057]  |
| <i>X After 2007</i>                               | -0.113<br>[1.461]                         | 3.568<br>[2.067]  | 3.723<br>[2.74]   | 3.379<br>[3.721]  |
| Bid-ask spread, in bps                            | 0.004<br>[0.007]                          | 0.004<br>[0.01]   | -0.008<br>[0.013] | -0.031<br>[0.018] |
| <i>X After 2007</i>                               | -0.025<br>[0.013]                         | -0.003<br>[0.018] | 0.027<br>[0.024]  | 0.069<br>[0.032]  |
| Implied volatility                                | 0.022<br>[0.079]                          | 0.101<br>[0.112]  | 0.250<br>[0.149]  | 0.039<br>[0.202]  |
| <i>X After 2007</i>                               | 0.098<br>[0.092]                          | 0.003<br>[0.13]   | -0.119<br>[0.173] | 0.036<br>[0.235]  |
| Number of observations                            | 15666                                     | 15604             | 15511             | 15340             |
| R-squared                                         | 0.413                                     | 0.444             | 0.462             | 0.480             |
| Adjusted R-squared                                | 0.389                                     | 0.421             | 0.440             | 0.459             |
| RMSE                                              | 1.801                                     | 2.548             | 3.378             | 4.586             |

Note: Standard errors are robust with clustering at the currency level. Numbers in green shade are statistically significant at the 95 percent level (two-sided test) and higher levels of the variable support currency in risk-off episode. Orange background denotes risk-off depreciation. Sample coverage is the same as in Table 5.



**Fig. 4.** Median of EM currency return betas with VIX and AUDJPY. Note: Betas are the estimated coefficients in a regression of a currency's weekly USD returns on the weekly return to the VIX or AUDJPY exchange rate using a sample of 52 weekly observations. Source: Bloomberg and authors' calculations.

While there is heterogeneity in the cross-sectional behavior of exchange rates across risk-off events, the following patterns stand out. Right after the onset of the risk-off episode, high yield currencies and those whose returns have a higher beta with respect to the AUDJPY exchange rate tend to have larger depreciations relative to the U.S. dollar. Stronger current account balances and net foreign asset positions are factors related to smaller risk-off depreciations or larger appreciations. While the increased frequency of risk-off episodes indicates that global factors have become more important, we would emphasize that country-specific factors remain essential in determining the impact of those global factors on specific currencies.

Looking at individual currencies, the Swiss franc and the Japanese yen tend to appreciate relative to the U.S. dollar during risk-off episodes. Because most other currencies depreciate relative to the U.S. dollar during risk-off episodes, the Swiss franc and the Japanese yen appreciate even more on a trade-weighted basis. On the other hand, a wide range of currencies, including those of Australia, Brazil, Canada, Chile, Colombia, India, Indonesia, Korea, Malaysia, Mexico, New Zealand, Peru, Philippines, Singapore, South Africa, Sweden, and Turkey, tend to depreciate during risk-off episodes, for at least some window.

Our findings could be helpful to policymakers. Knowing which currencies are most likely to appreciate or depreciate in future risk-off episodes is useful information for policymakers interested in the consequences of risk-off episodes for monetary policy and asset management. In countries with safe haven currencies, risk-off episodes bring on challenges related to rapid currency appreciation, such as loss of export competitiveness and deflation risks (for the Swiss case, see [de Carvalho Filho, 2015](#)). In countries whose currencies are perceived as relatively risky, risk-off episodes may cause FX weakness, with possible consequences for inflation, and may be disruptive in some situations, if there is a high degree of “dollarization” of liabilities.

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